CAYHIC 2026

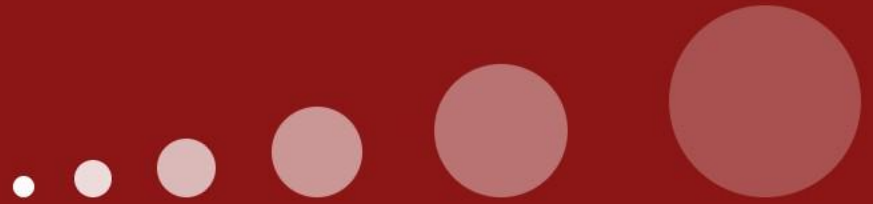
From Gene to Medicine

Retracing a cancer drug's discovery with AlphaFold and free genomics tools

Ulugbek Baymuradov

Computational Biologist & AI Engineer

Slides and every tool: **github.com/Bek/gene-to-medicine**



RIGHT NOW

A race to build a virtual cell

Switch **one gene** off in a cell.

Predict how all the other genes react.

5,000+
people signed up in 2025

114
countries

$100k
top prize (half cash)

What do you need to enter?

A lab | A PhD | ✓ A laptop

Free data. Free Kaggle GPUs. Age 18+.

2ND PLACE, 2025

One student. Working alone.

Public data. A simple neural network. A $50,000 prize.

SO WHY PAY FOR A VIRTUAL CELL?

Making medicines is slow and risky

10+

years to make one medicine

9 of 10

drugs that enter human trials fail

Let's follow one real medicine: HER2 → Herceptin → Enhertu

ABOUT ME

Hi, I'm Ulugbek

- Grew up in Uzbekistan
- Stanford: helped build the ENCODE portal
- Berkeley Lab · New York Genome Center
- Worked from a boat in the Pacific
- **Now: AI and data for cancer drugs**

I'm a computational biologist.

Tonight you'll meet people like me at every step.

BIOLOGY IN 4 MINUTES

The chain of life

DNA the ERBB2 gene → **RNA** a working copy → **Protein** HER2 → **Cell** a breast cell → **Tissue** a tumor → **Person** the patient

A protein's shape decides its job. What you can see or measure is the **phenotype**.

COMPUTERS NOW PREDICT EACH STEP

AlphaFold → the shape **Virtual cell → the cell** **Digital twin → the patient**

BIOLOGY IN 4 MINUTES

How cancer works

Cancer = cells that divide when they should not.

Oncogene

A stuck accelerator

Tumor suppressor

Broken brakes

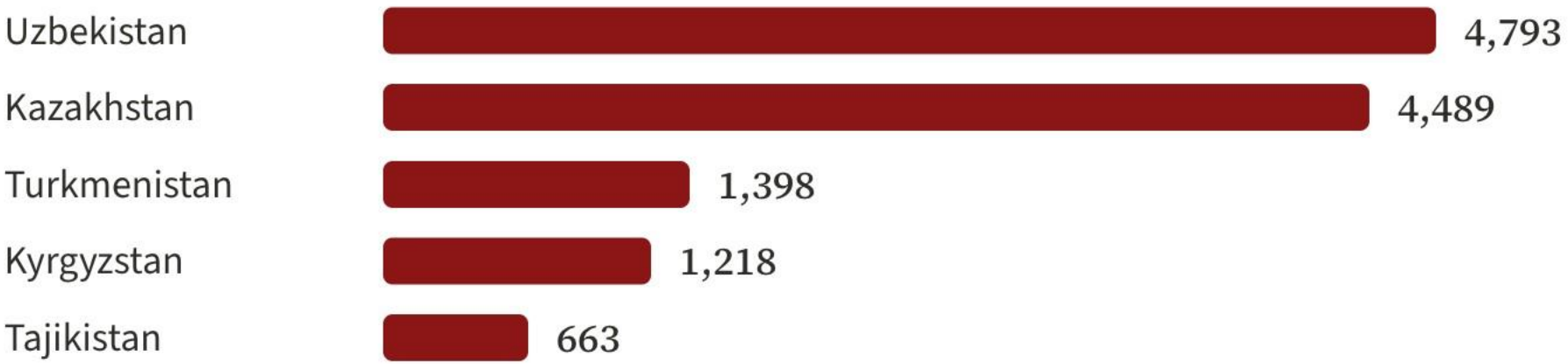
Is HER2 an accelerator or a brake?

Accelerator. A cancer cell can carry up to 2 million HER2 proteins (normal: 20,000).

WHY BREAST CANCER?

The #1 cancer in women here

New cases per year · #1 in women in all 5 Central Asian countries



About 1 in 5 is HER2-positive: our story tonight.

Global Cancer Observatory (IARC), 2024 estimates · Try it: gco.iarc.who.int → Fact sheets → your country

The gene hunter and the doctor

Axel Ullrich

1985: copied the HER2 gene

Dennis Slamon

1987: tested 189 frozen tumors

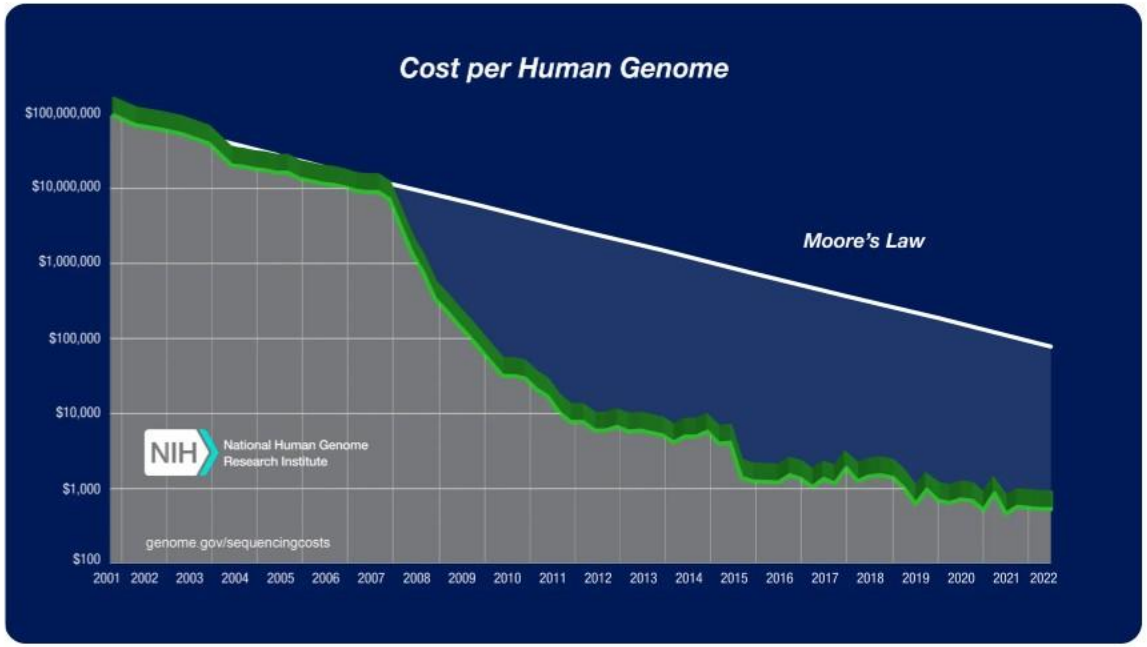
“Would you be interested in a collaboration?”

Slamon to Ullrich, over dinner

Guess: what % of the tumors had extra HER2 copies?

30%. And those women relapsed and died sooner. (Science, 1987)

Reading DNA got 180,000× cheaper



$95M

one genome, 2001

$525

one genome, 2022

So scientists read **11,000+ tumors**, and put them online for free.

Chart: NHGRI, genome.gov/sequencingcosts (Wetterstrand KA)

Try it: cBioPortal



14%

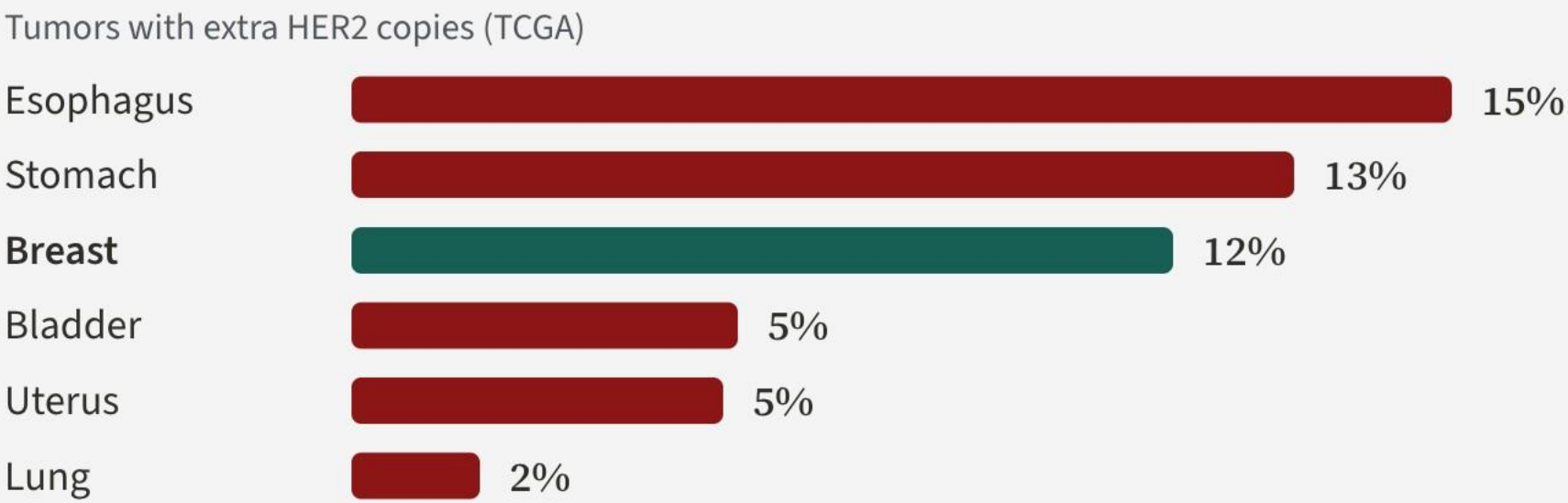
of 996 tumors have HER2 changes, mostly extra copies. Slamon 1987: 30%. Hospital tests: 15–20%. Different tests, different numbers.

Slamon needed years. You need 30 seconds.

Try it: cbioportal.org → Breast (TCGA, PanCancer Atlas) → gene ERBB2 → Submit

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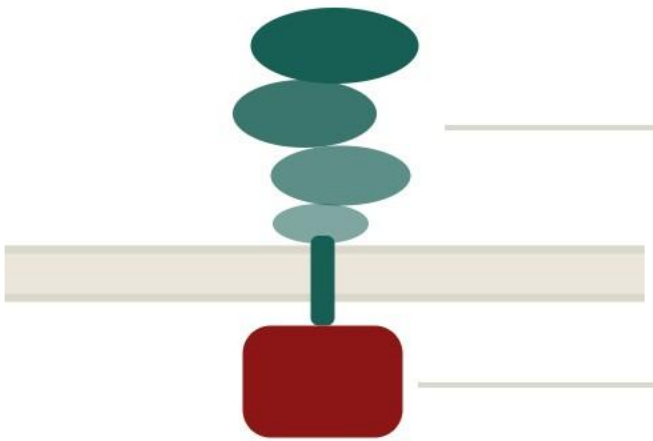
HER2 is not only in breast cancer



Remember the stomach. It comes back.

Try it: cBioPortal → all TCGA PanCancer studies → ERBB2 → Cancer Types Summary

HER2 is an antenna on the cell



Outside — catches signals

Membrane — the cell's outer skin

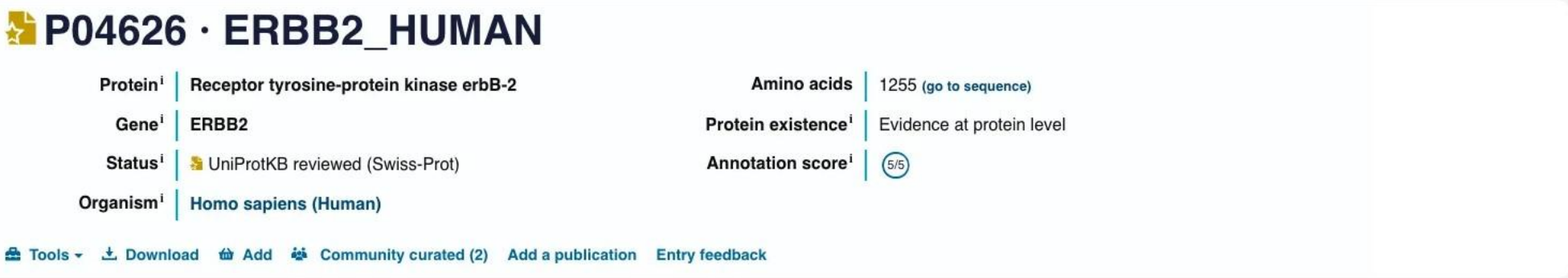
Inside — the switch that says “grow”

Where would you aim a drug: outside or inside?

Both work. Our drug grabs the outside. Pills block the inside.

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Try it: UniProt, the protein's ID card



1,255
amino acids long

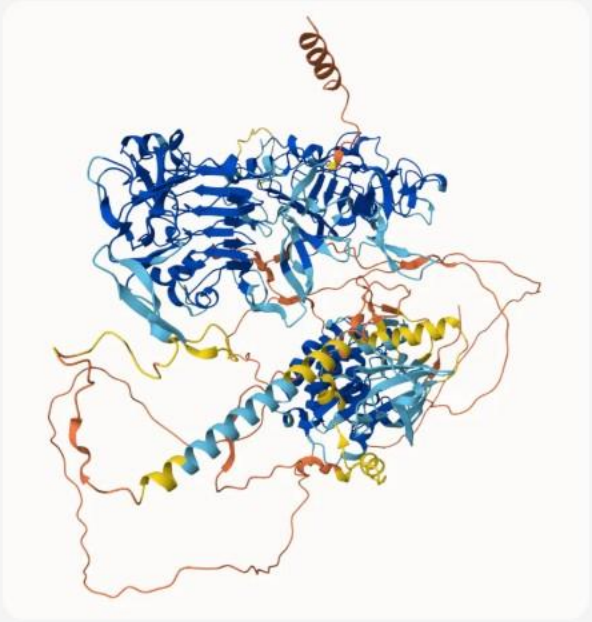
Membrane
where it lives

P04626
its ID in every tool

Try it: uniprot.org → search “HER2 human” → P04626

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AlphaFold: a 50-year problem, solved



Blue = sure · orange = unsure

About 260,000

protein shapes solved in labs since 1958

200 million+

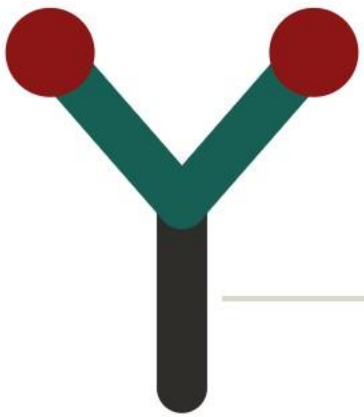
predicted by AlphaFold, free for everyone

Nobel Prize 2024

for its makers

Try it: alphafold.ebi.ac.uk → search P04626

An antibody is a Y-shaped grabber



Tips — grab one exact target

Stem — calls the immune system

1989: Genentech makes a **mouse** antibody against HER2.

Why not inject the mouse antibody into patients?

Our immune system would attack it as foreign.

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The drug almost died

Should the company keep going?

“For a period of time, the answer… was: No.”

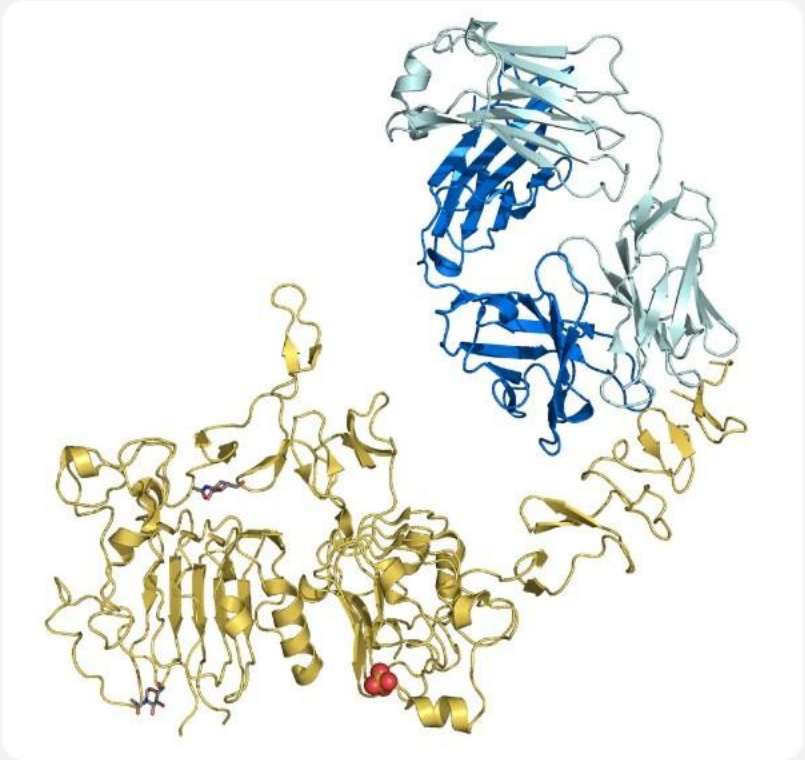
Genentech's own history

Bill Young
His mother had breast cancer

Mike Shepard
Kept a small team going

1992: mouse parts swapped for human ones (“humanized”) → trastuzumab (Herceptin)

See it in 3D: the drug grips HER2



- **Blue:** trastuzumab, the drug
- **Gold:** HER2, the target

1992

Humanizing one antibody took a team years.

2025

AI designs antibodies from scratch (first lab results).

Try it: rcsb.org → 1N8Z → 3D View

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Switch a gene off. Watch the cell.



Most cancer cells: **keep growing**



HER2-heavy cancer cells: **often stop**

DepMap switched off every gene, one by one, in **1,000+ cancer cell lines**.

Same idea as the Virtual Cell Challenge: switch a gene off, predict what happens.

Try it: depmap.org → search ERBB2

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Virtual cells, virtual mice

Cell: close

Free models exist, not yet accurate. That's the $100k race.

Mouse: not yet

An open problem for your generation.

Patient: early

“Digital twins” help some trials.

FDA, 2025: fewer animal tests, starting with antibody drugs

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Clinical trials: 100 drugs in, 8 out



After approval: Phase 4, watching for rare side effects.

Which phase fails the most drugs?

Phase 2: about 7 in 10 drugs fail there.

Source: BIO / Informa / QLS, clinical success rates 2011–2020

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Barbara Bradfield, 1992

A teacher. Her cancer had spread to her neck and lung.

She almost chose an unproven clinic. Slamon called her again the next morning.

April 1993: no sign of cancer.

Of the first 15 women, she was the only survivor.

A lottery for a medicine

1994: Dr. Marti Nelson asked for the new drug. Genentech said no.

She died at 40. Patients protested.

1995: places given out by lottery.

25

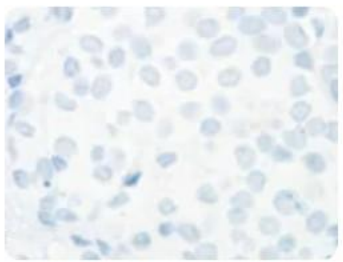
places every 3 months

Is a lottery a fair way to share a new drug?

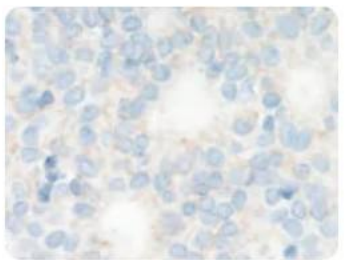
No perfect answer. Today we have “expanded access” programs.

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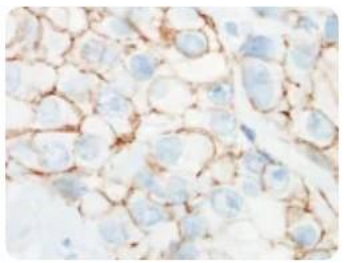
Test first, then treat



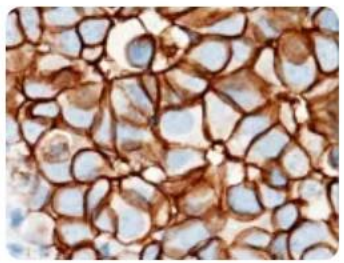
Score 0



1+



2+



3+ positive

469, not 8,000+

patients needed, by testing first

5 months

longer life in the big trial

1998

drug and test approved together

What if the trial gave the drug to ALL breast cancer patients?

Most would not benefit. The drug would look like a failure.

Stain images: Nitta et al. 2012, CC BY 2.5 · Trial: Slamon et al., NEJM 2001

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A surprise in the big trial

Which organ had the surprise?

The heart.

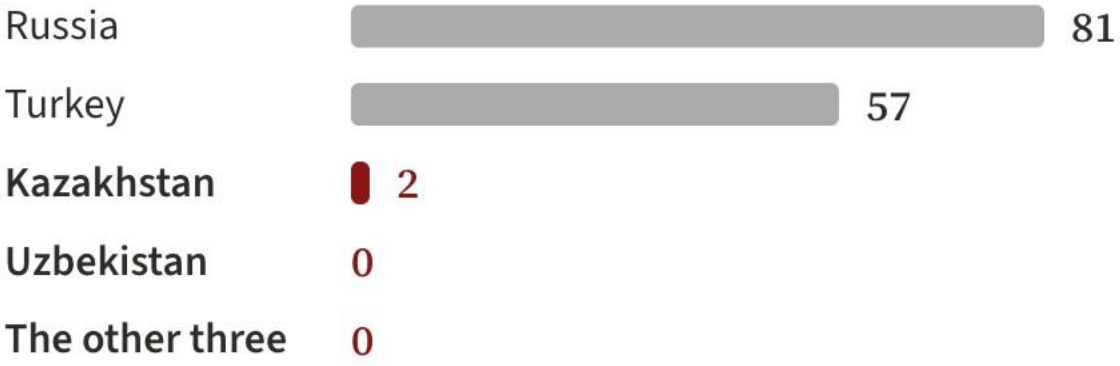
27% had heart problems with one type of chemo plus trastuzumab (8% with that chemo alone).
Now every patient gets heart scans.

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Try it: ClinicalTrials.gov

1,735

trastuzumab studies worldwide



Guess: how many list a site in your country?

Almost none. A gap, not a verdict.

Try it: clinicaltrials.gov → Intervention: trastuzumab → Location: your country (checked 27 Sep 2026)

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Reaching more patients

2005

After surgery: cancer comes back half as often

2015

On the WHO list of essential medicines

2017

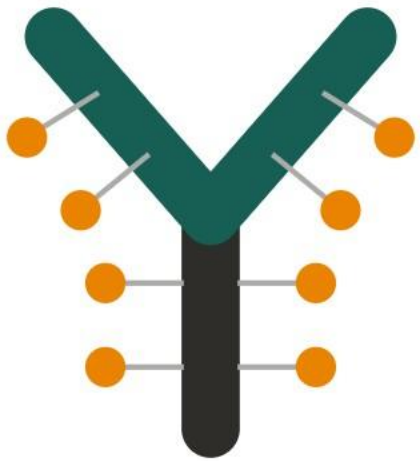
Europe and US approve cheaper copies

2019

WHO certifies its first cheaper copy: this drug

Cheaper copies (“biosimilars”) cost about 65% less.

Next generation: a guided missile



ADC = antibody-drug conjugate: an antibody that carries a drug.

- **Antibody:** finds HER2
- **Chemo:** the warhead, about 8 per antibody

ASCO 2022, THE BIGGEST CANCER MEETING

It worked even with a little HER2. Doctors stood and clapped.

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Enhertu, 2019 → 2026

Year	Approved for
2019	HER2-positive breast cancer that has spread
2021	Stomach cancer
2022	Breast cancer with only a little HER2
2024	Any solid tumor with a lot of HER2 (remember the stomach?)
Dec 2025	First treatment for HER2-positive breast cancer that has spread
May 2026	Early breast cancer, before and after surgery

US FDA approvals · sources: FDA; AstraZeneca and Daiichi Sankyo

THE WHOLE JOURNEY

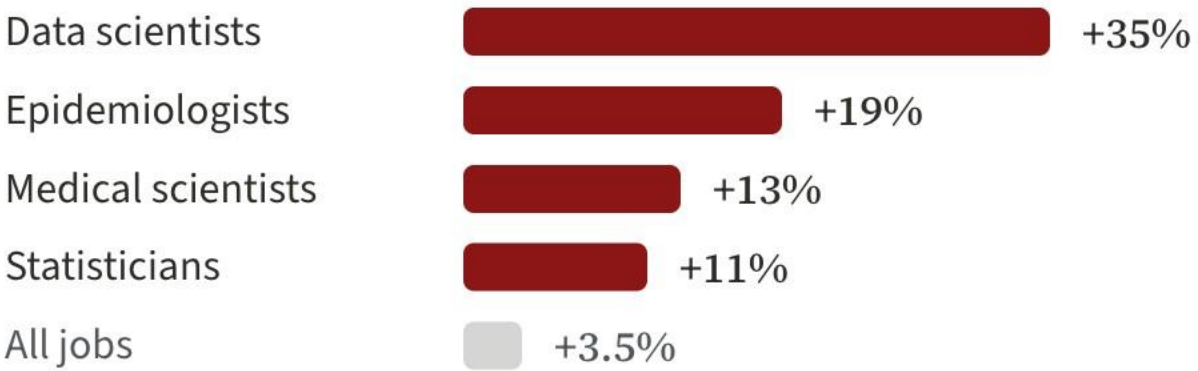
Who works at every step

Step	Free tool	Computers predict	Jobs
Gene	cBioPortal	which genes changed	pathologist, genomic data scientist
Protein	UniProt, AlphaFold	the protein's shape	structural biologist, AI engineer
Drug	PDB 3D viewer	new antibodies	antibody engineer, manufacturing
Cell	DepMap	how cells respond	pharmacologist, toxicologist
Patient	ClinicalTrials.gov	who will benefit	trial coordinator, biostatistician
World	Cancer Observatory	who needs the drug	epidemiologist, health economist

WHERE JOBS ARE GROWING

People who speak biology and data

US job growth, 2025–2035



CAN BE DONE REMOTELY

- Clinical data manager
- Statistical programmer
- Drug-safety specialist
- Bioinformatics analyst

Close to home: Uzbekistan's **“1000 Genomes”** project is reading its own people's DNA.

US Bureau of Labor Statistics, projections 2025–2035. A direction, not a promise.

OPEN NOW · AGE 18+ · OUR COUNTRIES CAN ENTER

Enter with a laptop

DEADLINE 5 NOV

Virtual Cell Challenge

Predict how cells react when a gene is switched off

$100k

top prize, half cash

virtualcellchallenge.org

DEADLINE 2 DEC

Virtual Embryo Challenge

Predict how an embryo's cells change as it grows

$8k

top prize in each track

virtualembryo.ai

DEADLINE 14 DEC

Identify the molecule

Predict which molecule a lab machine measured

$16k

1st place ($50k in total)

kaggle.com → “Enveda CASMI”

Small teams can win. Last year's 2nd place: one student.

UNDER 18?

Your path starts now

Virtual Embryo Challenge

Join a team with an adult captain. Write a helpful tutorial: up to $200.

Olympiads

International Olympiad in AI · International Biology Olympiad

Science fairs

Regeneron ISEF, through your national science fair (Kazakhstan: DARYN)

Breakthrough Junior Challenge

Ages 13–18. A 2-minute science video. $250k scholarship. Next round ~May 2027.

2026: an 11th-grader from Taraz reached ISEF with an AI project built only on a computer.

HOW TO BE READY

Four steps, starting tonight

1. Try the free tools on one gene you care about
2. Enter one competition, even with a simple first model
3. Write one page about what you found
4. Email a lab head or program coordinator with that page

THANK YOU

Your turn.

5 November. Last year, one student working alone came second.

Slides, every tool and every competition:

github.com/Bek/gene-to-medicine

Questions? Type them in the chat.

